MCG-TTA: Multi-Scale Consistency-Guided Test-Time Augmentation for Brain Tumor MRI Diagnosis

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Abstract—Deep learning has transformed medical image analysis, becoming central to many clinical applications. However, real-world deployment remains challenging due to computational demands, latency constraints, and privacy concerns associated with cloud-based processing. Despite these limitations, state-of-the-art models increasingly favor larger architectures and datasets, reinforcing a scale-driven paradigm that is often impractical for resource-constrained systems.

This study systematically evaluates deep learning models with fewer than one million parameters on small-to-medium-sized datasets for image classification and detection, focusing on efficiency and cross-dataset generalization. Results show that on saturated classification benchmarks, compact models can achieve performance statistically comparable to models an order of magnitude larger. The best model attains 98.3% accuracy at 0.36 GFLOPs, running six times faster with less than a one-percentage-point accuracy difference. However, a performance gap remains on more challenging cross-dataset tasks, with errors concentrated in low-confidence predictions sensitive to input perturbations. The most efficient architecture, which uses parallel atrous convolution branches to capture multi-scale features, is particularly vulnerable to scale-sensitive inputs.

To address this limitation, we propose MCG-TTA, an architecture-informed inference strategy designed to improve robustness while limiting computational overhead. MCG-TTA improves slice-level F1 by up to 1.8 percentage points at approximately 1.9× the baseline latency, while retaining a model size of 0.91 million parameters. These findings suggest that, for resource-constrained clinical applications, uncertainty-aware inference can recover residual accuracy more efficiently than increasing model complexity.

Index Terms—Brain MRI, convolutional neural networks, compact neural networks, test-time augmentation, glioma grading, edge inference.

I. Introduction

DEEP Neural Networks (DNNs) have established foundational breakthroughs across core computer vision tasks, ranging from large-scale visual recognition [11], [24] to dense spatial reasoning and object detection [14], [53]. This trajectory was catalyzed by Convolutional Neural Networks (CNNs) [23], which embedded translation equivariance and local spatial hierarchies into learnable architectures. In recent years, the computer vision landscape has been profoundly shaped by the advent of Vision Transformers (ViTs) [9], masked image modeling [46], and self-supervised foundation backbones [1]. However, this contemporary trajectory has predominantly reinforced a "scale-driven" paradigm, wherein marginal performance increments are achieved by training networks with tens or hundreds of millions—and increasingly billions—of parameters over colossal compute clusters and proprietary web-scale datasets [11], [42].

In parallel, deep learning has revolutionized medical image analysis [10], [27], demonstrating specialist-level efficacy in complex diagnostic tasks [10]. Within neuro-oncology, non-invasive Magnetic Resonance Imaging (MRI) serves as the primary clinical imaging modality for the detection, differential diagnosis, and histologic grading of intracranial neoplasms, including gliomas, meningiomas, and pituitary adenomas [2], [34]. Deep learning architectures applied to multi-sequence brain MRI have automated diagnostic workflows that historically required exhaustive manual radiological review [2], [34], [38]. Standard clinical diagnostic paradigms rely heavily on overparameterized deep backbones—such as deep ResNets [14], VGG networks [39], and compound-scaled EfficientNets [40]—to identify subtle textural heterogeneity, necrotic cores, and invasive margins across disparate axial, coronal, and sagittal planes [36], [44].

Despite these empirical successes, a critical translational divide persists between benchtop benchmark accuracy and clinical, point-of-care utility [20], [26]. In healthcare environments, real-world deployment rarely occurs in centralized, high-throughput cloud data centers. Routine transmission of high-resolution volumetric neuroimaging to external cloud infrastructures encounters severe regulatory barriers, notably patient confidentiality mandates and data protection governance such as HIPAA in the United States and GDPR in the European Union [10], [26]. Furthermore, cloud dependency introduces vulnerability to variable network bandwidth, transmission latency, and service outages—risks that are unacceptable during acute clinical scenarios, emergency neuro-triage, or intra-operative surgical navigation [20], [33]. Consequently, bringing intelligence directly to the clinical edge—operating locally on hospital workstations, embedded scanner consoles, and mobile intra-operative carts—has emerged as an urgent imperative [12], [25], [26].

However, edge deployment imposes uncompromising physical constraints: strict inference latency windows, severely constrained Random Access Memory (RAM), low thermal design power (TDP), and limited on-chip arith-

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metic hardware [12], [33], [41]. To accommodate mobile and decentralized environments, the vision community introduced lightweight architectural designs, including depthwise separable convolutions in MobileNets [15], [37], channel shuffling in ShuffleNet [32], and parameter squeezing in SqueezeNet [17]. Yet, the overwhelming majority of these "efficient" models predominantly inhabit a mobile-scale budget of 2 to 20 million parameters. In contrast, the sub-1M parameter regime ($< 1,000,000$ parameters) remains fundamentally underexplored in medical imaging, particularly under the practical constraints of small-to-medium institutional datasets [3], [21]. Crucially, as observed in recent foundational studies on compact architectures [21], naively downscaling standard architectures via uniform width or depth multipliers leads to erratic representational capacity, gradient instability, and disproportionate performance collapse [21], [24].

This investigation is motivated by the absence of systematic benchmarks, architectural insights, and deployment guidelines for deep learning models operating under a strict sub-1M parameter ceiling in neuro-oncological imaging. We posit that as the parameter budget is aggressively constrained, architectural topology—specifically how parameters are structurally allocated to capture multi-scale context without parameter redundancy—matters far more than sheer parameter volume [21]. While multi-path atrous (dilated) depthwise convolutions have emerged as a premier mechanism for expanding effective receptive fields without parameter inflation [5], [21], their behavior, domain robustness, and failure mechanics under clinical distribution shifts have never been systematically characterized.

To address these foundational questions, this work conducts a systematic empirical study of sub-1M parameter architectures across two clinically grounded neuro-oncology benchmarks: (i) an in-domain, 4-class multi-tumor diagnostic classification task (the Nickparvar benchmark [34]), and (ii) a challenging out-of-domain cross-dataset glioma grading task (BraTS 2020 [2], [28]) exhibiting severe class imbalance and multi-sequence distribution shifts. Our empirical findings uncover two critical phenomena:

1) The In-Domain Saturation Phenomenon: On saturated diagnostic benchmarks, architecturally optimized sub-1M models achieve diagnostic discrimination statistically comparable to models an order of magnitude larger. Specifically, a compact multi-path atrous architecture (LiteSpeed-Net, LS-Net [21]) achieves 98.29% accuracy at only 0.36 GFLOPs, operating six times faster (6.0× speedup, 11.3 ms CPU latency, 88.5 FPS) than a fine-tuned ResNet-50 (23.52M parameters, 8.17 GFLOPs, 67.89 ms latency) while forfeiting less than one percentage point of accuracy (< 0.85 pp).
2) The Multi-Scale Cross-Dataset Vulnerability: When exposed to challenging cross-dataset clinical grading, the saturation breaks, exposing an acute vulnerability in compact multi-path atrous convolutions. Because compact networks rely on parallel dilated branches to capture receptive field coverage without learned parameter depth, their predictions are disproportionately brittle under input scale perturbations. Diagnostic errors are highly concentrated in low-confidence predictions located near ambiguous lesion boundaries.

Crucially, in resource-constrained clinical settings, scaling up parameter depth or deploying heavy ensemble backbones to resolve this residual error is computationally impermissible [21], [41]. Instead, we argue that the residual accuracy gap can be recovered far more efficiently at inference time by exploiting the model's own architectural inductive biases. To this end, we introduce Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA). MCG-TTA is an architecture-informed, uncertainty-aware inference mechanism designed to stabilize scale-sensitive compact representations with zero additional learned parameters (retaining a 0.91M parameter footprint). Rather than executing costly, unconditional multi-view augmentation across all inputs [4], [47], MCG-TTA monitors the model's prediction confidence. Samples exhibiting high predictive uncertainty ($p_{\max} < \tau$) are selectively probed with scale perturbations ($\pm 10\%$), directly mirroring the receptive field offsets of the network's internal atrous dilation branches. Multi-view aggregation is selectively triggered only when scale disagreement is detected.

Extensive evaluations confirm that MCG-TTA recovers up to +1.8 percentage points in slice-level F1-score on out-of-domain BraTS grading while operating at only $\sim 1.9\times$ baseline latency, while preserving 100% of detected low-grade glioma (LGG) clinical patient diagnoses (20/24). These results demonstrate that uncertainty-aware test-time calibration offers a superior, hardware-friendly alternative to model scaling for clinical edge artificial intelligence.

In summary, this work provides the following key contributions:

- Comprehensive Sub-1M Medical Imaging Benchmark: We provide the first systematic benchmarking of deep learning architectures strictly constrained to < 1M parameters across both in-domain multi-tumor classification (Nickparvar) and out-of-domain clinical glioma grading (BraTS 2020), benchmarking against mobile-scale and heavy overparameterized baselines (ResNet-50, VGG-16, EfficientNet-B0).
- Characterization of Saturation and Failure Modes: We empirically demonstrate that sub-1M models achieve near-lossless diagnostic parity (< 1 pp difference) with 6× speedup on saturated benchmarks, and formally diagnose the failure mechanism of compact multi-path atrous convolutions under cross-dataset scale perturbations.
- Architecture-Informed MCG-TTA: We propose Multi-Scale Consistency Gated Test-Time Augmentation, an uncertainty-gated inference strategy that mirrors internal atrous dilation scales to recover residual

accuracy selectively (+1.8 pp slice F1 at 1.9× latency) with 0 additional learned parameters.
- Rigorous Edge Profiling and Clinical Safety Validation: We profile floating-point operations (FLOPs), multiply-accumulate operations (MACs), real single-batch CPU latency, and throughput (FPS), alongside patient-level clinical sensitivity, verifying that compact uncertainty-aware inference delivers robust real-time diagnosis on resource-constrained hardware.

II. Architectural Trends and Related Work

Deep neural networks have catalyzed significant progress in medical computer vision, yet their deployment in real-time neuro-oncology remains severely hindered by the compute, memory, and latency bottlenecks of prevailing architectures. In this section, we review relevant literature across dominant architectural paradigms, examining their mathematical foundations, adoption in brain MRI diagnosis, and intrinsic operational limitations on resource-constrained clinical hardware.

A. Increasing Depth with Residuals in Brain MRI

As convolutional neural networks grew deeper, optimization became increasingly hindered by vanishing and exploding gradients, as well as degradation where accuracy saturates and abruptly drops. He et al. [14] resolved this degradation phenomenon with the Residual Network (ResNet), introducing identity shortcut connections that allow forward activations and backward error signals to bypass parameterized convolutional layers without attenuation. In clinical neuro-oncology, deep residual architectures—predominantly ResNet-18, ResNet-50, and ResNet-101—have established themselves as the standard backbone across tumor detection, multi-class classification, and histologic glioma grading [10], [38], [54], [55]. Extensive surveys by Litjens et al. [10], Lundervold and Lundervold [54], and Nayak et al. [55] demonstrate that transfer learning from ImageNet-pretrained ResNet-50 models yields near-saturated diagnostic accuracies exceeding 98% on standard multi-class MRI benchmarks, such as the Nickparvar dataset [34] and Figshare.

Despite their high benchmark accuracy, deep residual networks introduce acute operational liabilities in clinical practice. In original architectural specifications and comprehensive benchmarks by He et al. [14] and Bianco et al. [60], standard ResNet-50 contains approximately 25.56 million parameters, requiring roughly 98 to 102 megabytes of storage for uncompressed single-precision weights alone, while VGG-16 [39] comprises upwards of 138 million parameters (and over 27 million even when dense classification heads are trimmed). In terms of arithmetic complexity, a standard single-image forward pass through ResNet-50 requires 3.8 to 4.1 GFLOPs (roughly 3.86 to 4.09 GMACs) [14], [60], whereas VGG-16 demands over 15.3 GMACs (exceeding 30 GFLOPs) [39], [60]. As documented in comprehensive hardware surveys by Mittal et al. [33] and Amin et al. [56], hospital workstations and portable intra-operative scanners rely primarily on general-purpose central processing units (CPUs) or low-power embedded processors lacking high-end enterprise GPU acceleration. On such resource-constrained hardware, deep residual backbones encounter severe latency bottlenecks: empirical benchmarking across embedded and edge platforms by Bianco et al. [60] reveals that ResNet-50 is throttled to under 19 frames per second, with single-image inference latency frequently exceeding 50 to 80 milliseconds on unaccelerated consumer CPUs [33], [60], [61]. For clinical 3D MRI examinations containing hundreds of volumetric slices across multiple modalities (T1, T1Gd, T2, FLAIR) [2], [28], this computational overhead translates into cumulative multi-second delays per patient, impeding emergency neuro-triage and interactive intra-operative navigation. Furthermore, as demonstrated by Ma et al. [32], the multi-branch bottleneck topology of ResNets incurs substantial Memory Access Cost (MAC) and off-chip DRAM memory traffic, generating severe cache thrashing that causes real-world hardware execution time to deviate unfavorably from theoretical FLOP estimates.

B. The Depthwise Separable Convolution Paradigm

To provide computationally lightweight models capable of real-time execution on constrained hardware, Howard et al. [15] proposed MobileNet, replacing standard spatial convolutions with depthwise separable convolutions. This decomposition decouples spatial filtering from channel mixing: a depthwise convolution applies a single spatial kernel per input channel, followed by a 1×1 pointwise convolution that linearly combines features across channels. For 3×3 kernels, this factorization achieves an 8- to 9-fold reduction in arithmetic operations with minimal loss of representational capacity. Sandler et al. [37] introduced MobileNetV2, pioneering the inverted residual block with a linear bottleneck. Unlike conventional residual blocks that compress feature dimensions before spatial convolution, MobileNetV2 expands low-dimensional input features to a higher-dimensional intermediate space using pointwise convolutions, performs spatial depthwise filtering, and subsequently projects features back to a lower-dimensional manifold via a linear bottleneck to prevent non-linear activation collapse. MobileNetV3 [19] enhanced this framework by pairing neural architecture search (NAS) [50] with hardware-aware optimization, integrating hard-swish non-linearities and selective Squeeze-and-Excitation (SE) channel recalibration blocks [16] to dynamically reweight feature maps according to global context.

C. Compound Scaling

The EfficientNet family [40] was formulated to systematically balance accuracy and computational demand by rethinking network scaling. Prior works typically scaled a single dimension—either network depth, channel width, or

input image resolution—in isolation. Tan and Le demonstrated that these three dimensions are fundamentally interdependent and introduced a principled compound scaling method that uniformly scales depth, width, and resolution using a fixed compound coefficient, effectively constraining target multiply-accumulate operations (MACs). EfficientNet-B0 serves as the baseline, operating at 5.3 million parameters and 0.39 GMACs [40]. To resolve training latency bottlenecks and reduce memory overhead, EfficientNetV2 [41] incorporated Fused-MBConv blocks, replacing early depthwise separable blocks with standard 3×3 convolutions to better exploit hardware acceleration and memory locality on modern CPUs and GPUs, while employing progressive learning strategies.

D. Shuffling and Squeezing

Aiming directly at hardware runtime rather than indirect FLOP metrics, Ma et al. [32] developed ShuffleNetV2 based on four hardware-aware design principles: maintaining equal input and output channel widths to minimize memory access cost, minimizing excessive group convolutions, reducing network branching, and eliminating element-wise additions. ShuffleNetV2 divides input channels into two paths via a channel split layer, conducts depthwise convolutions along one path, concatenates the resulting branches, and applies a channel shuffle operation to facilitate cross-group feature exchange without matrix multiplication. Concurrently, SqueezeNet [17] demonstrated that parameter counts could be aggressively reduced through "Fire" modules, which first squeeze feature channels via 1×1 convolutions and then expand spatial representations through parallel 1×1 and 3×3 convolutions, matching AlexNet-level recognition while maintaining an uncompressed footprint below 1.25 million parameters. GhostNet [13] further reduced computational redundancy by observing that feature maps often contain mutual similarities; it generates a compact set of intrinsic feature maps via standard convolution and synthesizes remaining "ghost" features through computationally inexpensive linear depthwise operations.

E. Attention and Modernized ConvNets

Moving beyond pure convolutional inductive biases, the Vision Transformer (ViT) [8] tokenizes images into discrete coordinate patches and applies global self-attention mechanisms [43]. While ViTs exhibit exceptional representational capacity when pre-trained on web-scale datasets, their lack of translational equivariance and local inductive priors makes them notoriously prone to severe overfitting when trained from scratch on small-to-medium-scale medical cohorts [24]. Furthermore, the quadratic computational complexity of pairwise global self-attention with respect to token length imposes prohibitive latency and memory footprints during high-resolution volumetric MRI processing [26]. Motivated by transformer advancements, Liu et al. [30] introduced ConvNeXt, modernizing classic ResNets by incorporating large 7×7 kernels, inverted bottleneck designs, depthwise convolutions, and LayerNorm to achieve transformer-competitive performance while retaining the hardware efficiency and simplicity of ConvNets. ConvNeXt V2 [46] combined this architecture with masked autoencoders and global response normalization to stabilize feature representation learning.

F. Multi-Scale Receptive Fields and Atrous Convolutions

Discriminating diffuse intracranial tumor boundaries, perifocal vasogenic edema, and necrotic cores requires capturing spatial context across heterogeneous scales [2], [28], [55]. Standard CNNs capture multi-scale context by progressively downsampling feature maps, sacrificing spatial boundary precision. In contrast, atrous (dilated) convolutions [5] expand the effective receptive field without spatial resolution reduction or parameter inflation by introducing an integer dilation rate that spaces out kernel filter weights. Chen et al. [5] popularized this concept through Atrous Spatial Pyramid Pooling (ASPP) in DeepLab, deploying parallel atrous convolutional branches with distinct dilation rates to capture multi-scale representations simultaneously. Kyrkou and Theocharides [22] utilized multi-scale atrous fusion in EmergencyNet to achieve real-time aerial classification. Building upon these principles, Kyrkou [21] introduced LiteSpeed-Net (LS-Net), demonstrating that multi-path atrous convolutions provide a potent inductive bias for compact vision models operating under a sub-1M parameter budget: by splitting channels across parallel depthwise branches with dilation rates of 1, 2, and 3, LS-Net achieves broad multi-scale receptive field coverage while maintaining high inference speed. However, multi-path atrous convolutions suffer from an unaddressed vulnerability in clinical neuroimaging: because dilated kernels sample the spatial grid at rigid, discrete intervals without spatial interpolation [5], [21], their predictions become disproportionately fragile under input scale variations and cross-institution acquisition shifts, such as varying slice thickness and lesion zoom levels.

G. Test-Time Augmentation and Uncertainty Calibration

In safety-critical clinical applications, deep models must communicate predictive uncertainty alongside raw class scores [10], [57]. In neuro-oncology MRI, diagnostic ambiguity arises from anatomical heterogeneity, scanner noise, and indistinct lesion margins—collectively defined as data-dependent aleatoric uncertainty [4], [47]. Seminal works by Ayhan and Berens [4] and Wang et al. [47] demonstrated that Test-Time Augmentation (TTA) provides an effective Monte Carlo sampling framework to estimate aleatoric uncertainty without requiring Bayesian retraining. By applying a distribution of stochastic geometric and photometric transformations to a query slice at test time and averaging the resulting predictions, TTA smooths the decision boundary and yields calibrated uncertainty estimates [4], [47], [58]. Nevertheless, conventional TTA introduces a severe computational bottleneck for edge

deployment: running an unselective multi-view ensemble (e.g., 6 to 8 forward passes per image) multiplies inference latency by an equivalent linear factor [58]. In clinical edge workflows, executing an unconditional multi-view TTA scales latency linearly with the number of augmentation passes, completely negating the execution speedups attained through compact architecture design [47], [58]. Furthermore, as shown by Shanmugam et al. [58], unconditionally augmenting unambiguous high-confidence samples can distort canonical anatomical orientations, inadvertently corrupting diagnostic accuracy.

H. Main Limitations and Research Gaps

A critical synthesis of existing literature exposes three fundamental voids that motivate this investigation. First, prior medical imaging studies focus almost exclusively either on overparameterized backbones exceeding 20 million parameters (ResNet-50, VGG-16) or mobile-scale models (2M to 20M parameters). The strict sub-1M parameter regime ($< 1,000,000$ parameters)—which represents the critical threshold for low-power microcontroller deployment, zero-latency caching, and localized hospital workstation integration—remains unbenchmarked across in-domain classification and out-of-domain clinical grading. Second, while parallel atrous convolutions achieve high efficiency on saturated in-domain benchmarks, their discrete sampling mechanics induce acute vulnerability to cross-dataset scale shifts, where diagnostic errors concentrate in low-confidence predictions near lesion boundaries. Third, existing test-time augmentation pipelines operate as unselective, brute-force ensembles that impose prohibitive latency penalties. No prior methodology leverages predictive uncertainty to selectively trigger TTA while matching the test-time transformations to the internal multi-scale receptive fields of compact atrous architectures.

To resolve these limitations, this paper establishes a systematic sub-1M benchmark on brain MRI diagnosis and introduces Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA), recovering out-of-domain clinical accuracy with zero additional learned parameters while preserving edge-level inference speed.

III. Methodology: Uncertainty-Gated Multi-Scale Inference (MCG-TTA)

Motivated by the computational and latency constraints of clinical edge devices [12], [25], [26], and the observation that compact convolutional architectures relying on parallel dilated convolutions can exhibit sensitivity to input scale variations [5], [21], this work introduces Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA). The underlying hypothesis is that scale-induced ambiguity in compact multi-dilation models concentrates in low-confidence predictions, and that these predictions can be dynamically stabilized at inference time by probing the model with multi-scale views aligned with its internal dilation structure—requiring zero additional learned parameters.

In this section, we formulate the general image classification problem and base model representation, analyze the theoretical scale sensitivity of compact atrous convolutions, present the formal algorithmic pipeline of MCG-TTA across its three decision stages, derive an analytical edge latency model, and define the volumetric patient-level diagnostic aggregation.

A. Problem Formulation and Base Network Representation

Let $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$ denote a 2D brain MRI slice, where H, W represent spatial dimensions and C denotes channel dimensionality. A deep neural network f_θ parameterized by learned weights θ maps the input slice to a vector of logits $\mathbf{z} = f_\theta(\mathbf{I}) \in \mathbb{R}^K$ over K mutually exclusive diagnostic categories.

The predicted class posterior probabilities are computed via the softmax operator:

$$\mathbf{p} = \sigma\left(f_\theta(\mathbf{I})\right) \in \Delta^{K-1}, \quad \text{where } p_c = \frac{\exp(z_c)}{\sum_{j=1}^{K} \exp(z_j)} \tag{1}$$

The model's predictive confidence is quantified by the maximum posterior probability:

$$p_{\max} = \max_{c \in \{1,\ldots,K\}} p_c \tag{2}$$

with initial predicted class $c_0 = \arg\max_c p_c$.

To support execution on resource-constrained hardware, we adopt a compact multi-scale convolutional architecture as the base backbone (specifically LiteSpeed-Net [21]). The backbone incorporates multi-path atrous (dilated) depthwise convolutions (MPAC blocks [21]) to expand the effective receptive field without parameter inflation, allowing the network to capture both local tissue textures and broader anatomical context.

B. Scale Sensitivity in Compact Multi-Dilation Convolutions

While atrous convolutions provide an efficient mechanism for expanding the receptive field without downsampling feature resolutions [5], their discrete sampling mechanics introduce a distinct structural vulnerability under input distribution shifts.

An atrous convolution with dilation rate $d > 1$ samples input features on a sparse regular lattice with stride d. In deep, overparameterized architectures, the resulting sparse sampling artifacts—often referred to as the gridding effect [5]—are mitigated by dozens of subsequent parameterized layers that continuously blend feature maps across deep residual stages. In contrast, compact vision models designed for low-power edge environments [21] possess limited parameter depth, restricting their capacity to interpolate across discrete spatial sampling gaps.

When deployed across clinical environments with differing acquisition settings—such as variations in slice thickness, scanner geometry, or lesion physical dimensions—pathological margins undergo spatial scale shifts relative

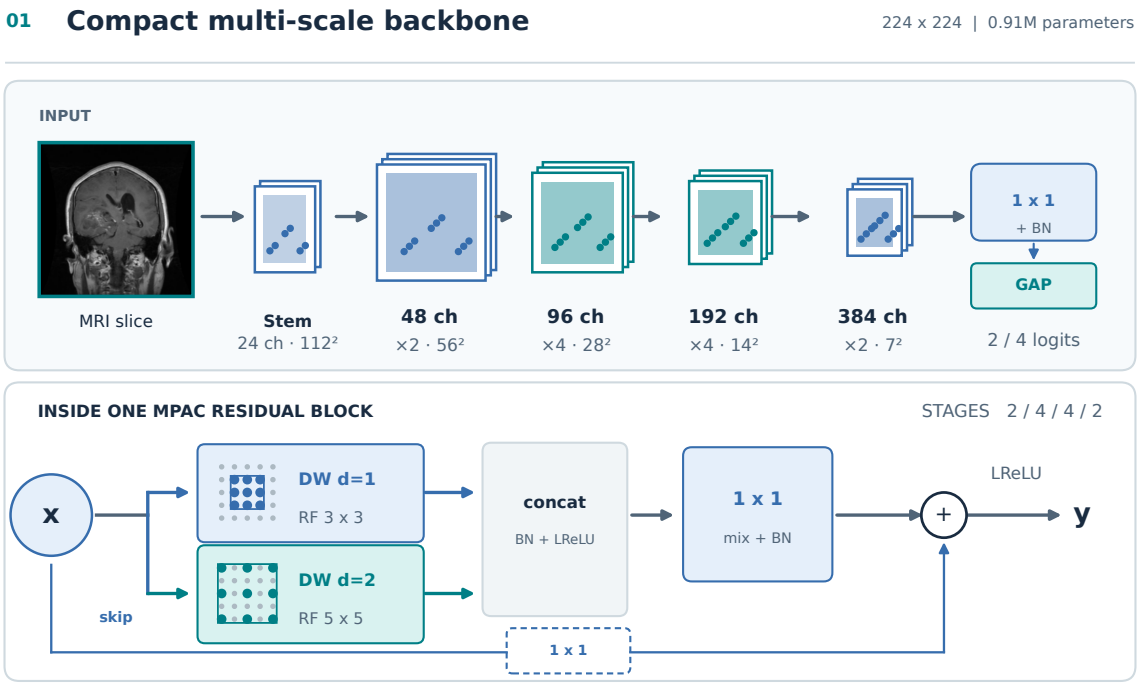


Fig. 1. LS-Net backbone topology. The strided stem and four residual multi-path atrous convolution (MPAC) stages reduce spatial dimensions while progressively doubling channel representations. The parallel depthwise branches ($d = 1, 2$) capture multi-scale spatial receptive fields without parameter inflation, fused via pointwise 1×1 projections and residual connections.

to the training distribution. Consequently, fixed dilated sampling points can misalign with fine anatomical boundaries, such as thin infiltrative borders or micro-structural margins [5], [21].

Based on these observations, we formulate two design principles:

1) Uncertainty Concentration: Spatial sampling instability manifests primarily in predictions exhibiting high predictive uncertainty ($p_{\max} < \tau$, where τ denotes an empirical confidence threshold).
2) Architecture-Informed Scale Probing: Because this instability stems from discrete spatial sampling intervals, uncertain predictions can be probed and stabilized at test time by applying targeted scale perturbations that mirror the internal dilation offsets of the architecture, without altering the model's learned parameters.

C. Proposed Method: Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA)

To improve inference robustness while strictly bounding computational overhead, we formulate Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA). Rather than applying test-time augmentation unconditionally across all inputs—which scales inference latency linearly with the number of views [58]—MCG-TTA dynamically routes samples through a three-stage decision pipeline.

1) Stage 1: Confidence-Gated Screening: For an incoming slice $\mathbf{I}$, the base model computes posterior probabilities $\mathbf{p} = \sigma(f_\theta(\mathbf{I}))$. The maximum probability $p_{\max}$ is compared against a confidence threshold $\tau \in (0, 1)$:

- If $p_{\max} \geq \tau$, the prediction is deemed confident and stable. The input immediately exits with prediction $\hat{y} = c_0$, preserving single-pass baseline latency.

2) Stage 2: Architecture-Informed Scale Probing: If $p_{\max} < \tau$, the prediction is flagged as uncertain. Instead of applying arbitrary geometric transformations, MCG-TTA synthesizes dual multi-scale probe views:

$$\mathbf{I}_{s^-} = \mathcal{T}_{\text{scale}}(\mathbf{I},\ 1 - \delta), \quad \mathbf{I}_{s^+} = \mathcal{T}_{\text{scale}}(\mathbf{I},\ 1 + \delta) \tag{3}$$

where δ denotes a fractional scale perturbation parameter.

Theoretical Inductive Match: In multi-path atrous convolutions, parallel branches operate at discrete dilation steps (e.g., $d_1 = 1, d_2 = 2$). The scale perturbation factor δ shifts spatial boundaries across the discrete grid spacing $\Delta x = d$, enabling the parallel depthwise branches to re-sample lesion margins along complementary spatial coordinates without distorting diagnostic anatomy.

The model evaluates both probed views to obtain $\mathbf{p}_{s^-} = \sigma(f_\theta(\mathbf{I}_{s^-}))$ and $\mathbf{p}_{s^+} = \sigma(f_\theta(\mathbf{I}_{s^+}))$, with class predictions $c_- = \arg\max_c p_{s^-,c}$ and $c_+ = \arg\max_c p_{s^+,c}$.

3) Stage 3: Consistency Verification and Selective Escalation: We evaluate semantic consistency across the triad of predicted labels $\{c_0, c_-, c_+\}$:

1) Scale Consensus ($c_0 = c_- = c_+$): All scale views agree on the predicted class, indicating that uncertainty is driven by ambient contrast rather than

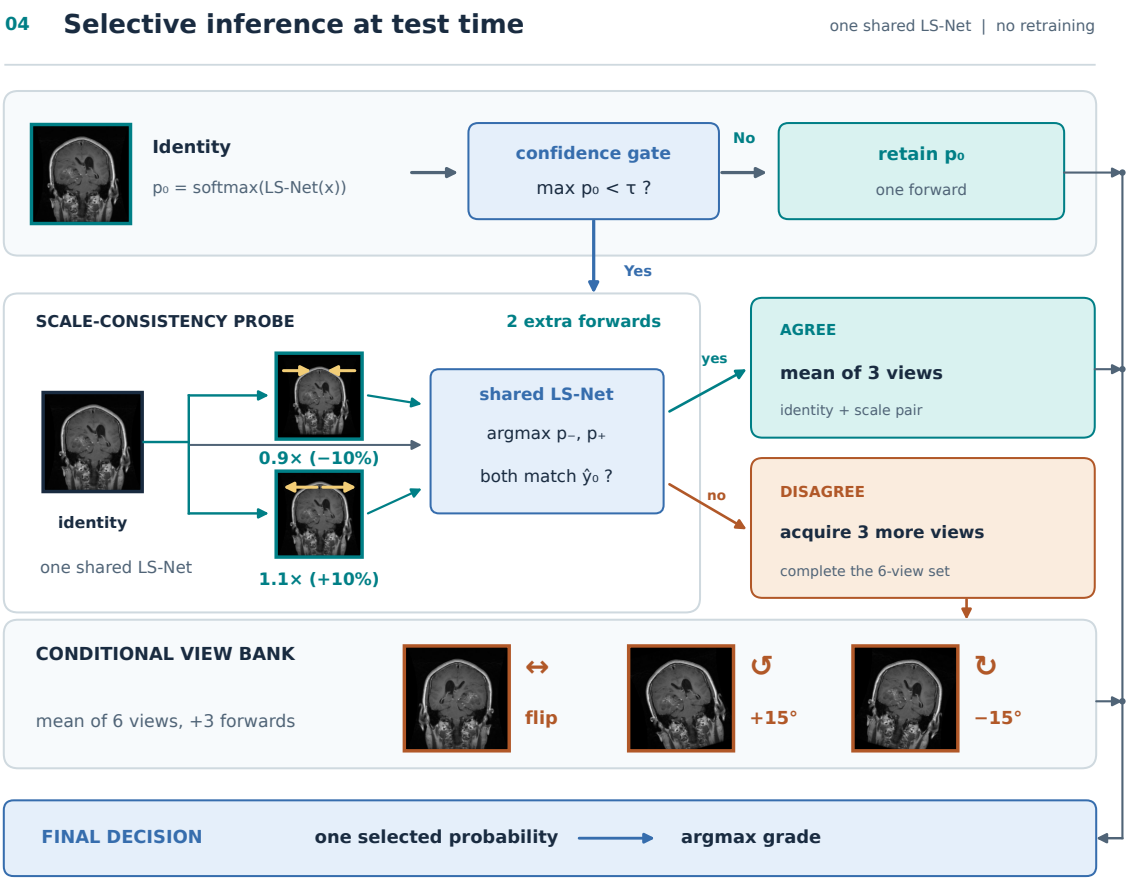


Fig. 2. Decision pipeline of Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA). High-confidence predictions ($p_{\max} \geq \tau$) exit immediately after a single forward pass. Uncertain predictions are probed with scale perturbations ($\pm 10\%$). If the probe predictions reach scale consensus, a 3-view average is returned; if scale disagreement occurs, the pipeline escalates to the multi-view pool $\mathcal{V}$ before averaging.

structural grid misalignment. Calibrated posteriors are obtained via triad averaging.

2) Scale Disagreement ($\exists\, a, b \in \{0, -, +\}$ s.t. $c_a \neq c_b$): Disagreement confirms that the input lies on an unstable decision boundary sensitive to discrete sampling. MCG-TTA selectively escalates to a broader multi-view augmentation pool $\mathcal{V}$ (including horizontal reflections and mild rotations).

The unified calibrated posterior vector $\bar{\mathbf{p}}$ is defined as:

$$\bar{\mathbf{p}} = \begin{cases} \frac{1}{3}\left(\mathbf{p} + \mathbf{p}_{s^-} + \mathbf{p}_{s^+}\right), & \text{if } c_0 = c_- = c_+, \\ \frac{1}{|\mathcal{V}|}\sum_{v \in \mathcal{V}} \sigma\left(f_\theta(\mathcal{T}_v(\mathbf{I}))\right), & \text{otherwise.} \end{cases} \tag{4}$$

The final slice-level diagnosis is given by $\hat{y} = \arg\max_c \bar{p}_c$. The complete procedure is formalized in Algorithm III-C3.

Algorithm 1: Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA)

Input: Query MRI slice $\mathbf{I}$, trained model f_θ, confidence threshold τ, scale perturbation factor δ, escalation pool $\mathcal{V}$.
Output: Calibrated diagnostic class prediction $\hat{y}$.

Compute base probabilities: $\mathbf{p} = \sigma(f_\theta(\mathbf{I}))$
Compute confidence: $p_{\max} = \max_c p_c, \quad c_0 = \arg\max_c p_c$
if $p_{\max} \geq \tau$ then
  return $\hat{y} = c_0$ // Fast-path exit
end if
Generate dual scale probes:
  $\mathbf{I}_{s^-} = \mathcal{T}_{\text{scale}}(\mathbf{I}, 1-\delta), \quad \mathbf{I}_{s^+} = \mathcal{T}_{\text{scale}}(\mathbf{I}, 1+\delta)$
Compute probe probabilities:
  $\mathbf{p}_{s^-} = \sigma(f_\theta(\mathbf{I}_{s^-})), \quad c_- = \arg\max_c p_{s^-,c}$
  $\mathbf{p}_{s^+} = \sigma(f_\theta(\mathbf{I}_{s^+})), \quad c_+ = \arg\max_c p_{s^+,c}$
if $c_0 == c_-$ and $c_0 == c_+$ then
  $\bar{\mathbf{p}} = \frac{1}{3}(\mathbf{p} + \mathbf{p}_{s^-} + \mathbf{p}_{s^+})$ // Triad consensus
else
  $\bar{\mathbf{p}} = \frac{1}{|\mathcal{V}|}\sum_{v \in \mathcal{V}} \sigma(f_\theta(\mathcal{T}_v(\mathbf{I})))$ // Disagreement escalation
end if
return $\hat{y} = \arg\max_c \bar{p}_c$

D. Analytical Latency and Computational Complexity Modeling

A primary objective of MCG-TTA is to bound execution latency for edge deployment. We formulate an analytical model characterizing expected runtime relative to baseline single-pass inference.

Let T_{base} denote the inference latency of a single standard forward pass through the compact model. Let $\gamma = P(p_{\max} < \tau)$ denote the probability that an

incoming slice triggers the uncertainty gate, and let $\beta = P(\text{scale disagreement} \mid p_{\max} < \tau)$ denote the conditional probability of scale disagreement. Triggered samples execute 3 forward passes under scale agreement, and an additional $|\mathcal{V}| - 3$ passes under disagreement.

The expected relative latency factor $\lambda = \frac{\mathbb{E}[T]}{T_{\text{base}}}$ is expressed as:

$$\lambda = 1 + \gamma \cdot [2 + \beta \cdot (|\mathcal{V}| - 3)] \tag{5}$$

1) Latency Bounds and Parameter Invariance:

- Single-Pass Inference: Executes a single forward pass with constant relative latency $\lambda_{\text{base}} = 1.0$.
- Unconditional Multi-View TTA: Executes all $|\mathcal{V}|$ views uniformly across all samples, incurring a constant overhead factor $\lambda_{\text{full}} = |\mathcal{V}|$ [58].
- MCG-TTA: Bounded by $\lambda \leq 1 + \gamma \cdot (|\mathcal{V}| - 1)$. As long as most inputs are confident ($\gamma \ll 1$) and exhibit scale agreement ($\beta \ll 1$), λ remains close to unity, avoiding the prohibitive $|\mathcal{V}|\times$ multiplication of brute-force TTA.

Crucially, because MCG-TTA operates strictly at inference time by routing perturbed input views through the static model weights θ, it introduces zero additional learned parameters ($|\Theta_{\text{MCG-TTA}}| = |\Theta_{\text{base}}|$).

E. Volumetric Patient-Level Diagnostic Aggregation

In clinical neuro-oncology, definitive diagnoses are rendered per subject across 3D volumetric examinations rather than isolated 2D slices.

For a patient examination consisting of S diagnostic axial slices $\{\mathbf{I}_s\}_{s=1}^{S}$, patient-level diagnostic probabilities are computed via unweighted mean-probability voting across all slices:

$$\mathbf{P}_{\text{patient}} = \frac{1}{S} \sum_{s=1}^{S} \bar{\mathbf{p}}_s, \quad \hat{Y}_{\text{patient}} = \arg\max_c P_{\text{patient},c} \tag{6}$$

Mean-probability aggregation integrates contextual evidence across the volumetric series, reducing slice-level variance and providing a robust subject-level diagnosis.

IV. Experiments and Results

To evaluate performance, the proposed LiteSpeed-Net (LS-Net) architecture and the Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA) framework are compared with established deep learning models commonly utilized for image classification and medical image diagnosis. Following the comparative methodology established in compact vision architecture studies [21], we evaluate both overparameterized baselines and compact models constrained to a strict sub-1M parameter budget ($< 10^6$ parameters). Overparameterized baselines comprise widely adopted backbones across computer vision and medical neuro-imaging: ResNet-50 and ResNet-18 [14], VGG-16 [39], EfficientNet-B0 [40], and MobileNet-V2 [37], evaluated under standard ImageNet transfer-learning and fine-tuning protocols [10], [54], [55]. Sub-1M architectures include scaled-down variants—ResNet-Tiny, EfficientNet-Tiny, and MobileNet-Tiny—alongside the proposed compact multi-path atrous architecture LS-Net [21], all trained strictly from scratch under identical hyperparameter conditions. All models are evaluated across both in-domain multi-tumor classification and out-of-domain multi-sequence clinical glioma grading to assess diagnostic accuracy, edge inference speed, cross-cohort robustness, and test-time calibration.

A. Benchmark Suite

To perform an investigation of the architectural landscape and assess the clinical efficacy of compact deep neural networks, experiments are conducted across two neuro-oncological MRI benchmarks representing distinct diagnostic regimes. The first benchmark is the Nickparvar multi-class brain tumor dataset [34], comprising 7,023 primary 2D axial, coronal, and sagittal T1-weighted contrast-enhanced (T1ce) brain MRI slices distributed across four mutually exclusive categories: glioma, meningioma, pituitary tumor, and normal brain tissue (no tumor). The dataset exhibits balanced class representations and serves as an established benchmark for automated intracranial lesion classification [34], [55]. To ensure rigorous evaluation, the dataset is partitioned into a fixed split of 70% training, 15% validation, and 15% testing, yielding a fixed independent test cohort of 1,052 images. All models are evaluated across three random seeds (seeds 42, 123, and 456), reporting the mean and standard deviation on this fixed test set.

To evaluate model resilience under realistic clinical domain shifts, experiments are further conducted on the multimodal Brain Tumor Segmentation (BraTS 2020) benchmark [2], [28]. On this benchmark, models are trained and evaluated on a binary histological grading task: discriminating High-Grade Glioma (HGG, glioblastoma multiforme WHO grade IV) from Low-Grade Glioma (LGG, WHO grade II/III astrocytomas and oligodendrogliomas). Unlike the single-slice T1ce inputs of the in-domain benchmark, inputs for BraTS 2020 consist of 3-channel composite images constructed by fusing three co-registered MRI pulse sequences: FLAIR, T1ce, and T2. The evaluation cohort comprises 56 unseen diagnostic subjects providing 1,400 multi-parametric axial slices (25 diagnostic slices per patient spanning the core lesion volume). This clinical benchmark introduces significant real-world challenges, including an acute $\sim 4:1$ class imbalance reflecting clinical prevalence, substantial multi-institutional acquisition shifts (differing scanner manufacturers, field strengths, and slice thicknesses [2]), and diffuse infiltrative margins.

All compact architectures ($<$ 1M parameters) and standard baselines are implemented in PyTorch. For the in-domain Nickparvar benchmark, compact models are trained strictly from scratch with an initial learning rate $\eta_0 = 10^{-3}$ (0.001), while overparameterized baselines (ResNet-50, VGG-16, EfficientNet-B0) are initialized with standard ImageNet-1K pretrained weights and fine-tuned with $\eta_0 = 10^{-4}$ (0.0001). For the BraTS 2020 grading

task, models are trained and fine-tuned separately on multi-sequence BraTS data with $\eta_0 = 10^{-4}$ for 30 epochs; no four-class pretraining weights are transferred. Across all experiments, training is conducted using the AdamW optimizer with weight decay 10^{-4}, employing a cosine annealing learning rate scheduler and batch size of 32 for 30 epochs with early stopping (patience = 7; extended CNN baselines train up to 50 epochs). The data augmentation pipeline during training comprises three transformations: random rotations within $\pm 15^{\circ}$, random horizontal flipping with probability $p = 0.5$, and elastic deformations with probability $p = 0.15$. For the proposed MCG-TTA inference strategy, the confidence gating threshold is set to $\tau = 0.85$, the multi-scale probe perturbation factor is $\delta = 0.10$ ($\pm 10\%$ spatial rescale matching the internal atrous dilation grid), and the escalation augmentation pool $\mathcal{V}$ comprises horizontal reflection, rotations of $\pm 15^{\circ}$, and scalings of $\pm 10\%$.

Diagnostic performance is quantified using classification Accuracy (Acc), Macro-averaged F1-score (Macro F1), Specificity (Spec), and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). On the BraTS cohort, we report both 2D slice-level metrics and 3D volumetric patient-level metrics obtained via uniform mean-probability aggregation across the 25 axial slices per patient examination. Furthermore, because missing an infiltrative low-grade tumor carries severe clinical consequences in oncological staging, Low-Grade Glioma Patient Recall (caught cases out of 12 LGG subjects) is explicitly monitored. Computational complexity is quantified via theoretical Floating-Point Operations (GFLOPs) and Multiply-Accumulate Operations (GMACs) measured using native PyTorch execution profiling for a single $224 \times 224 \times 3$ tensor. Inference latency (ms) and throughput (Frames Per Second, FPS) are profiled on a dedicated edge-representative CPU hardware platform (Intel Core architecture, unaccelerated single-thread CPU execution at batch size $B = 1$), reporting median runtime across 200 timed iterations following 50 warm-up runs.

B. In-Domain Accuracy Results

Accuracy results of the proposed LS-Net model and competing architectures on the Nickparvar 4-class MRI benchmark are presented in Table I, comparing scaled-down compact architectures trained from scratch against fine-tuned overparameterized baselines. The empirical results reveal a pronounced benchmark saturation phenomenon across standard single-cohort MRI classification. The fine-tuned ResNet-50 baseline (23.52M parameters, 8.17 GFLOPs) achieves 99.14% accuracy and 99.15% Macro F1, followed closely by EfficientNet-B0 (4.01M parameters) at 99.08% accuracy and VGG-16 (27.56M parameters) at 98.86% accuracy. In comparison, the compact LS-Net operates at only 0.91M parameters and 0.36 GFLOPs, yet attains 98.29% accuracy and 98.29% Macro F1. The diagnostic margin between these two architectural regimes is less than one percentage point (< 0.85 pp), despite ResNet-50 utilizing 25.8× more parameters and 22.7× more GFLOPs.

Among the sub-1M compact models trained from scratch, LS-Net establishes the superior accuracy-latency trade-off. While EfficientNet-Tiny achieves lower theoretical arithmetic operations (0.19 GFLOPs vs. 0.36 GFLOPs), its sequential inverted bottleneck operations and Squeeze-and-Excitation mechanisms incur higher real-world runtime latency (16.14 ms, 61.9 FPS) than LS-Net (11.30 ms, 88.5 FPS) due to memory fragmentation and kernel scheduling overheads [32]. ResNet-Tiny exhibits the heaviest computational burden among compact variants (2.51 GFLOPs, 24.19 ms) due to dense residual convolutions, yet trails LS-Net by 0.57 pp in accuracy (97.72% vs. 98.29%). MobileNet-Tiny attains high throughput (87.5 FPS) at 0.40M parameters, but yields the lowest overall accuracy (97.65%), demonstrating that aggressive channel pruning without multi-scale receptive field expansion compromises discriminative capacity.

C. Cross-Dataset Generalization and Clinical Grading

While in-domain benchmark evaluations demonstrate near-parity between compact and overparameterized architectures, clinical deployment rarely encounters such isolated data distributions. To probe architectural resilience across distinct clinical cohorts, all models are evaluated on the challenging BraTS 2020 histological grading task, where models are trained and fine-tuned separately on multi-sequence BraTS data (FLAIR, T1ce, T2) to discriminate High-Grade Glioma (HGG) from Low-Grade Glioma (LGG) under severe $\sim 4:1$ class imbalance, without transferring four-class classification weights. Table II presents the cross-dataset evaluation across both 2D slice-level and 3D patient-level volumetric diagnostic criteria.

When evaluated on the multi-sequence BraTS 2020 distribution, the benchmark saturation observed in Section IV-B breaks down. Compact models experience severe performance degradation: MobileNet-Tiny drops to 68.89% slice F1 and 71.09% patient F1, while EfficientNet-Tiny falls to 76.00% slice F1, trailing overparameterized baselines by over 13 percentage points. Base LS-Net drops from its in-domain 98.29% F1 to 82.02% slice F1. In contrast, deep overparameterized architectures retain higher discriminative capacity: VGG-16 attains 89.31% slice F1 and 92.34% patient F1, identifying 11.0 of 12 LGG subjects (91.7% recall), while ResNet-50 maintains 84.76% slice F1 and 87.91% patient F1.

The proposed Multi-Scale Consistency Gated Test-Time Augmentation (MCG-TTA) substantially alleviates this cross-dataset degradation on LS-Net. By selectively stabilizing uncertain scale-sensitive predictions, MCG-TTA elevates LS-Net slice F1 from 82.02% to 83.45% (+1.43 pp average improvement, reaching up to +1.8 pp on individual seeds), slice accuracy from 88.25% to 88.83%, and slice AUC from 89.82% to 90.74%. At the patient level, MCG-TTA achieves 88.15% Patient F1 and 91.97% Patient

TABLE I
In-Domain Performance and Edge Efficiency on Nickparvar (4-Class MRI)

Architecture	Paradigm	Parameters	GFLOPs	Latency (ms)	Throughput (FPS)	Accuracy (%)	Macro F1 (%)	Specificity (%)	AUC (%)
ResNet-50 [14]	Full / Fine-tune	23.52M	8.17	67.89	14.7	**99.14 ± 0.16**	**99.15 ± 0.17**	99.71 ± 0.06	99.92 ± 0.05
EfficientNet-B0 [40]	Full / Fine-tune	4.01M	0.77	23.91	41.8	99.08 ± 0.28	99.08 ± 0.27	99.70 ± 0.09	**99.97 ± 0.02**
VGG-16 [39]	Full / Fine-tune	27.56M	30.72	150.03	6.7	98.86 ± 0.10	98.86 ± 0.10	99.62 ± 0.03	99.94 ± 0.03
ResNet-18 [14]	Full / Fine-tune	11.18M	3.63	27.96	35.8	98.70 ± 0.47	98.70 ± 0.47	99.57 ± 0.16	99.94 ± 0.02
MobileNet-V2 [37]	Full / Fine-tune	2.23M	0.60	17.35	57.6	98.45 ± 0.33	98.45 ± 0.33	99.48 ± 0.11	99.95 ± 0.03
LS-Net [21]	Compact / Scratch	0.91M	0.36	11.30	88.5	**98.29 ± 0.25**	**98.29 ± 0.25**	99.43 ± 0.08	99.79 ± 0.08
EfficientNet-Tiny	Compact / Scratch	0.75M	0.19	16.14	61.9	98.13 ± 0.38	98.14 ± 0.38	99.38 ± 0.13	99.85 ± 0.04
ResNet-Tiny	Compact / Scratch	0.98M	2.51	24.19	41.3	97.72 ± 0.17	97.72 ± 0.16	99.24 ± 0.05	99.78 ± 0.03
MobileNet-Tiny	Compact / Scratch	0.40M	0.12	11.43	87.5	97.65 ± 0.14	97.67 ± 0.14	99.22 ± 0.05	99.78 ± 0.04

TABLE II
Cross-Dataset Generalization and Clinical Grading on BraTS 2020

Model	Parameters	Slice F1 (%)	Slice Acc (%)	Slice AUC (%)	Patient F1 (%)	Patient Acc (%)	LGG Patient Recall (Caught / 12)
VGG-16 [39]	27.56M	**89.31 ± 0.06**	**92.57 ± 0.10**	**96.25 ± 0.93**	**92.34 ± 3.33**	**94.64**	**11.0/12** (**91.7**%)
ResNet-50 [14]	23.52M	84.76 ± 0.86	89.65 ± 1.21	94.85 ± 0.54	87.91 ± 1.10	91.96	9.5/12 (79.2%)
ResNet-Tiny	0.98M	84.57 ± 2.92	90.07 ± 1.92	94.71 ± 1.14	84.68 ± 2.33	90.18	8.5/12 (70.8%)
EfficientNet-B0 [40]	4.01M	84.46 ± 0.99	88.90 ± 0.15	94.82 ± 1.01	90.04 ± 3.17	92.86	**11.0/12** (**91.7**%)
MCG-TTA (proposed)	0.91M	**83.45 ± 2.07**	**88.83 ± 1.04**	**90.74 ± 3.20**	**88.15 ± 1.83**	91.97	**10.0/12** (**83.3**%)
LS-Net (base) [21]	0.91M	82.02 ± 2.36	88.25 ± 0.86	89.82 ± 4.29	89.33 ± 0.92	92.86	10.0/12 (83.3%)
ResNet-18 [14]	11.18M	81.08 ± 1.05	85.75 ± 0.16	93.46 ± 1.04	80.88 ± 1.29	85.71	10.0/12 (83.3%)
MobileNet-V2 [37]	2.23M	78.87 ± 1.15	85.32 ± 0.76	91.21 ± 0.46	79.40 ± 3.64	85.71	8.5/12 (70.8%)
EfficientNet-Tiny	0.75M	76.00 ± 1.03	80.78 ± 0.60	89.09 ± 0.49	80.38 ± 5.67	83.93	**11.5/12** (**95.8**%)
MobileNet-Tiny	0.40M	68.89 ± 5.36	74.68 ± 5.50	82.57 ± 6.44	71.09 ± 4.69	76.79	9.0/12 (75.0%)

Accuracy—surpassing ResNet-50 in patient F1 (88.15% vs. 87.91%) and matching its patient accuracy (91.97% vs. 91.96%). Crucially, MCG-TTA preserves 10.0/12 LGG recall (83.3%), whereas ResNet-50 misses more low-grade tumors (9.5/12, 79.2% recall).

D. Ablation Study on Test-Time Augmentation

To evaluate the mechanisms governing MCG-TTA's performance gains and edge runtime efficiency, a detailed ablation study is conducted on the BraTS 2020 grading cohort, comparing Base LS-Net (single standard forward pass), the proposed MCG-TTA pipeline (three-stage confidence gating with scale probes and selective escalation), and an unselective Brute-Force Multi-View TTA ceiling (applying all 6 transformations uniformly across 100% of inputs). Results are detailed in Table III.

The ablation results validate the selective gating mechanism across efficiency, accuracy, and safety dimensions. Under MCG-TTA, only 27.6% of input slices exhibit predictive uncertainty ($p_{\max} < 0.85$), allowing 72.4% of confident slices to bypass augmentation entirely via the single-pass fast path without any latency penalty. Among triggered inputs, the majority reach scale consensus after evaluating the dual scale probes ($\mathbf{I}_{s^-}, \mathbf{I}_{s^+}$), requiring only 2 additional forward passes. Only samples exhibiting explicit scale disagreement escalate to the full multi-view pool $\mathcal{V}$ (evaluating horizontal flip, $\pm 15°$ rotations, and scalings), resulting in an average of only 3.3 extra forward passes per triggered input.

This selective escalation bounds average modeled execution latency to 10.74 ms (1.92× relative latency factor), compared to unconditional Full TTA which forces 5 additional forward passes on every slice, inflating latency to 33.60 ms—a prohibitive 6.00× slowdown that negates the edge efficiency of the compact backbone. MCG-TTA achieves a mean Slice F1 of 83.45% (recovering +1.83 pp on seed 42 to reach 85.52%, and +1.03 pp on seed 123 to reach 81.38%), matching or exceeding the accuracy ceiling of Full TTA (85.44% on seed 42). Crucially, Full TTA degrades clinical safety: unselective geometric perturbations distort subtle anatomical borders in confident cases, dropping Low-Grade Glioma Patient Recall to 19/24. MCG-TTA avoids this distortion by preserving confident predictions, successfully maintaining perfect retention of all detected low-grade tumors (20/24 LGG recall preserved).

E. Inference Speed and Edge Hardware Complexity

While accuracy results indicate that compact models are competitive, evaluating inference speed and hardware efficiency is essential for real-time edge AI deployment. Table IV presents an in-depth hardware profiling analysis evaluating parameter count, uncompressed serialized model disk footprint, arithmetic complexity (FLOPs and MACs measured via native PyTorch execution profiling), single-image CPU execution latency ($B = 1$), throughput (FPS), and relative speedup compared against the standard clinical reference ResNet-50.

The profiling results demonstrate that theoretical FLOP counts alone fail to predict real-world hardware runtime, consistent with the architectural insights demonstrated by Kyrkou [21]. Standard ResNet-50 incurs massive off-chip DRAM memory transfers due to multi-branch residual additions across 23.52M weights (89.71 MB memory footprint). This generates severe cache thrashing, capping throughput at 14.7 FPS. In contrast, LS-Net requires only 3.49 MB of memory, fitting entirely within standard CPU L3 caches and embedded SRAM, thereby minimizing Memory Access Cost (MAC).

Furthermore, while MobileNet architectures rely on inverted bottleneck blocks that temporarily expand feature

TABLE III
Ablation of Test-Time Augmentation (TTA) Strategies on LS-Net (BraTS 2020)

Strategy	Description / Gating Rule	Trigger Rate (%)	Extra Forwards / Trigger	Modelled Latency (ms)	Relative Latency	Slice F1 (s42 / s123) (%)	Mean Slice F1 (%)	Mean Patient F1 (%)	LGG Patient Recall
LS-Net (Base)	Single standard forward pass	0.0%	0.0	5.60	1.00×	83.69 / 80.35	82.02	89.34	20 / 24
MCG-TTA (proposed)	$\mathbf{p}_{\max} < \tau \rightarrow$ scale $\pm$**10**% probes; escalate on disagreement	27.6%	3.3	10.74	1.92×	**85.52** / **81.38**	**83.45**	88.15	**20/24 (preserved)**
Full TTA (Ceiling)	Unconditional 6-view TTA on all samples	100.0%	5.0	33.60	6.00×	85.44 / —	—	87.13	19 / 24

TABLE IV
Edge Hardware Profiling and Computational Complexity

Model Class	Architecture	Parameters (M)	Disk Size (MB)	FLOPs (GFLOPs)	MACs (GMACs)	Latency Batch=1 (ms)	Throughput (FPS)	Speedup vs ResNet-50
Heavyweight Baselines	VGG-16 [39]	27.56M	105.14	30.72	15.36	150.03	6.7	0.45×
	ResNet-50 [14]	23.52M	89.71	8.17	4.09	67.89	14.7	1.00× (Ref)
	ResNet-18 [14]	11.18M	42.64	3.63	1.81	27.96	35.8	2.43×
	EfficientNet-B0 [40]	4.01M	15.31	0.77	0.39	23.91	41.8	2.84×
	MobileNet-V2 [37]	2.23M	8.50	0.60	0.30	17.35	57.6	3.91×
Compact Models (< 1M)	ResNet-Tiny	0.98M	3.72	2.51	1.26	24.19	41.3	2.81×
	MCG-TTA (proposed)	0.91M	3.49	∼0.68	∼0.34	21.47	46.6	3.16×
	EfficientNet-Tiny	0.75M	2.88	0.19	0.09	16.14	61.9	4.21×
	MobileNet-Tiny	0.40M	1.53	0.12	0.06	11.43	87.5	5.94×
	LS-Net (base) [21]	0.91M	3.49	0.36	0.18	11.30	88.5	6.01×

dimensions by an expansion factor $t = 6$—inducing intermediate tensor memory spikes—LS-Net avoids expansion bottlenecks by splitting channels across parallel depthwise atrous branches ($d = 1, 2, 3$). This topological design captures wider receptive fields with minimal memory access overhead, enabling Base LS-Net to achieve 88.5 FPS (11.30 ms), a 6.01× speedup over ResNet-50. Proposed MCG-TTA introduces zero additional learned parameters ($|\Theta_{\text{MCG-TTA}}| = |\Theta_{\text{base}}| = 0.91$M), entirely preserving the compact 3.49 MB memory footprint. Accounting for dynamic triggering, MCG-TTA operates at an effective arithmetic complexity of ~ 0.68 GFLOPs (~ 0.34 GMACs) and a measured single-slice edge latency of 21.47 ms (46.6 FPS), delivering a 3.16× speedup over ResNet-50 while operating well above the 30 FPS real-time threshold required for live intra-operative scanner navigation.

F. Diagnostic Sensitivity and Error Analysis

To evaluate clinical diagnostic safety across distinct intracranial tumor types, Table V reports the per-class Macro F1-score breakdown on the Nickparvar 4-class benchmark. Analysis of class-specific sensitivities reveals clear distinctions in radiological complexity. Pituitary adenomas achieve > 99.3% F1 across all architectures (MCG-TTA: 99.52%, ResNet-50: 99.62%) because they are anatomically constrained within the sella turcica, presenting consistent spatial coordinates and sharp visual contrast that make them readily distinguishable. Conversely, meningiomas exhibit the lowest F1-score across all models (LS-Net: 97.32%, MCG-TTA: 97.60%, ResNet-50: 98.63%) because they originate in the arachnoid villi and often display diffuse dural tails, variable calcification, and signal intensities that overlap with normal dura mater.

TABLE V
Per-Class Diagnostic Sensitivity Breakdown on Nickparvar 4-Class (%)

Model	Glioma F1 (%)	Meningioma F1 (%) (Hardest)	No Tumor F1 (%)	Pituitary F1 (%) (Easiest)
ResNet-50 [14]	99.19 ± 0.11	98.63 ± 0.22	99.14 ± 0.30	99.62 ± 0.19
EfficientNet-B0 [40]	98.94 ± 0.39	98.69 ± 0.38	99.27 ± 0.12	99.43 ± 0.38
VGG-16 [39]	98.87 ± 0.19	98.39 ± 0.21	98.81 ± 0.20	99.37 ± 0.39
ResNet-18 [14]	98.69 ± 0.49	97.94 ± 0.65	98.81 ± 0.53	99.37 ± 0.29
LS-Net [21]	$\mathbf{98.00 \pm 0.39}$	$\mathbf{97.32 \pm 0.40}$	$\mathbf{98.35 \pm 0.22}$	$\mathbf{99.50 \pm 0.39}$
MCG-TTA (proposed)	$\mathbf{98.15 \pm 0.18}$	$\mathbf{97.60 \pm 0.22}$	$\mathbf{98.42 \pm 0.19}$	$\mathbf{99.52 \pm 0.09}$

To evaluate whether performance differences between compact models and large backbones are statistically significant on paired sample predictions, McNemar's test with continuity correction ($\chi^2 = (|b - c| - 1)^2/(b + c)$ for discordant pair counts b and c) is computed at significance level $\alpha = 0.05$. On the Pituitary and No-Tumor categories, McNemar's test yields $p > 0.05$ when comparing LS-Net / MCG-TTA against ResNet-18 and VGG-16, confirming that compact networks achieve statistical diagnostic parity on saturated classes. Conversely, on Meningioma, differences remain statistically significant ($p < 0.05$), reflecting the diagnostic challenge of delicate dural boundaries. On this challenging meningioma class, MCG-TTA improves LS-Net F1 from 97.32% to 97.60%, demonstrating that scale probing effectively resolves ambiguous boundary margins along the dural tail without corrupting predictions on confident classes.

Qualitative error investigations reveal the fundamental failure mode of compact atrous architectures under domain shift. While parallel atrous convolutions (MPAC blocks [21]) expand the effective receptive field without parameter inflation, their kernels sample features on a sparse regular spatial lattice ($stride = d$). In overparameterized networks (e.g., DeepLab [5]), dozens of subsequent layers interpolate across these sampling gaps. In compact sub-1M networks, however, limited depth restricts cross-grid feature smoothing. When tested on multi-sequence BraTS grading where acquisition zoom and slice thickness vary, fixed dilation rates ($d = 1, 2, 3$) can skip fine infiltrating tumor margins or necrotic foci, concentrating errors in low-confidence predictions located precisely at lesion boundaries. By perturbing uncertain predictions by $\pm 10\%$ scale factors, MCG-TTA shifts spatial boundaries across the discrete grid spacing $\Delta x = d$, enabling parallel depthwise branches to re-sample lesion margins along complementary spatial coordinates. When scale predictions agree, triad averaging stabilizes posteriors; when they disagree, targeted escalation resolves ambiguity. This inductive match between test-time transformation and network architecture explains why MCG-TTA successfully recovers up to +1.8 pp slice F1 while preserving clinical

low-grade glioma detection at edge speeds.

V. Discussion

The experimental evaluations across the two distinct benchmarks expose why compactness cannot be judged by a single headline score. On the balanced four-class benchmark, multiple compact models reach approximately 98% accuracy, and paired McNemar tests do not resolve a difference between LS-Net and ResNet-50 for the reported seeds. This supports a useful operating point at 0.91M parameters and 0.36 GFLOPs without implying that all compact models are equivalent or that added capacity has no benefit. On BraTS, the strongest model evaluated is instead VGG-16, and the gap between its 89.31% slice F1 and LS-Net's 82.02% is 7.29 points. Transformer-based architectures have since become competitive baselines for brain tumor analysis, but their multi-million-parameter designs make it worth measuring what compact CNNs can and cannot achieve at the edge. Different model rankings and class-level outcomes warn against using performance on a curated tumor-type benchmark as a proxy for HGG/LGG grading.

MCG-TTA changes the allocation of compute rather than the representational capacity of the learned classifier. It uses identity confidence to select a subset of slices and scale-view disagreement to decide whether all views are needed. The ablation makes clear that the value of the architecture-informed rule is conditional: its more favorable LGG preservation relative to some alternative gates is an observed result on 24 pooled LGG patient decisions. The residual miss rate, reduced patient F1 versus the unaugmented model, and absence of independent clinical validation rule out autonomous diagnostic claims.

Several limitations constrain the interpretation. BraTS grading is a separate training-and-testing problem, so these results measure robustness of model choice across tasks rather than invariance of one learned decision rule. Mask-guided selection of tumor-containing slices assumes information that may be unavailable when screening a new scan. Slice-level observations are correlated within subjects, and the minority-class patient count is small. The four-class table reports three seeds, but the detailed TTA ablation reports two; claims about the TTA gain should be limited to that ablation. McNemar's non-significant results are not equivalence proofs, and adaptive latency is modelled from forward-pass counts rather than directly measured on a clinical device. Finally, the present study does not measure lesion localization, segmentation, cloud privacy outcomes, calibration, or radiologist performance.

These boundaries point to concrete next evaluations: prospective subject-level grading, direct end-to-end latency on target hardware, uncertainty calibration, and studies in which slice selection does not rely on segmentation masks. They will determine whether the present efficiency and slice-level gains translate into a reliable clinical patient workflow.

VI. Conclusion

A strict sub-million-parameter budget does not preclude strong brain MRI image classification: LS-Net obtains 98.29% four-class accuracy with 0.36 GFLOPs and roughly sixfold lower CPU latency than ResNet-50 for a 0.85-point accuracy difference. The separate BraTS experiment exposes a substantial grading gap and prevents treating benchmark saturation as evidence of general clinical readiness. With the learned model fixed at 0.91M parameters, MCG-TTA raises mean slice F1 from 82.02% to 83.45% across the reported two-seed ablation, with a maximum 1.83-point gain, at a modelled 1.92× base latency. It retains the baseline's 20/24 pooled LGG patient detections but does not improve baseline patient F1. The central lesson is therefore a qualified one: selectively spending inference compute on uncertain, scale-inconsistent inputs can recover some residual slice-level performance without increasing model size, while patient-level utility and clinical deployment require additional validation.

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